



Image Formation

Chapter 2



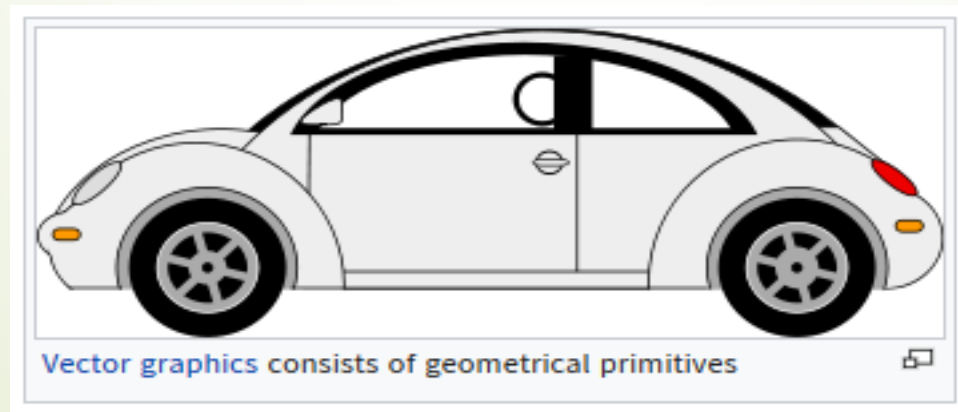
Contents

- ☆ Geometric primitives and Image Representation
 - ☆ 2D Geometric Transformations
 - ☆ 3D Geometric Transformations
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Geometric primitives and Image Representation

Geometric primitives or prim

- ★ Used in various senses, with the common meaning of the simplest (i.e. 'atomic' or irreducible) geometric objects that the system can handle (draw, store).
- ★ A common set of two-dimensional primitives includes lines, points, and polygons.
- ★ Some people prefer to consider triangles primitives because every polygon can be constructed from triangles.
- ★ All other graphic elements are built up from these primitives.



Geometric primitives and Image Representation

2D points

- ★ 2D points (pixel coordinates in an image) can be denoted using a pair of values,

$$x=(x,y) \in \mathbb{R}^2$$

$$x = \begin{bmatrix} x \\ y \end{bmatrix}$$

2D lines

- ★ 2D lines can be represented using homogenous coordinates

Line equation

$$ax + by + c = 0$$

Geometric primitives and Image Representation

Image as Functions

- ★ We think of an image as a function, f or I from R^2 to R :
 $f(x,y)$ gives the intensity or value at (x,y)
- ★ Practically define the image over a rectangle, with a finite range:
 $f:[a,b] \times [c,d] \rightarrow [\min, \max]$

2D Geometric Transformations

Transformations

- ☆ What is transformations?
 - ✓ The geometrical changes of an object form a current state to modified state.
- ☆ Why the transformations is needed?
 - ✓ To manipulate the initially created object and to display the modified object.



2D Geometric Transformations

Types of Transformations

★ Basic Transformations:

- ✓ Translation
- ✓ Scaling
- ✓ Rotation

★ Purposes:

- ✓ To move the position of objects
- ✓ To alter the shape/ size of objects
- ✓ To change the orientation of objects

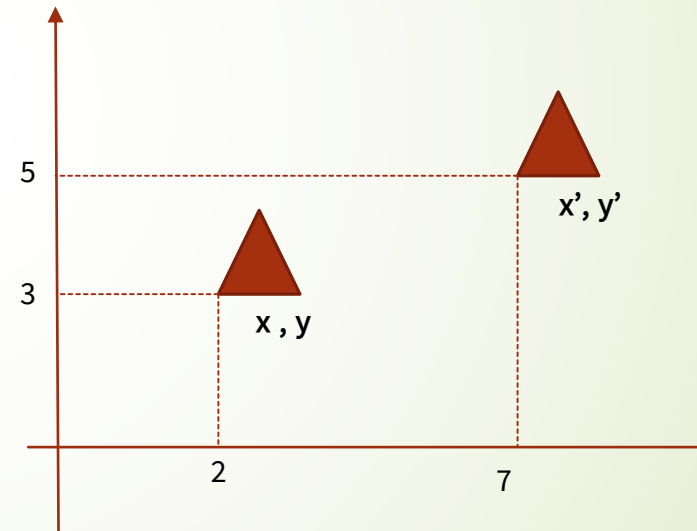
2D Geometric Transformations

Translation

- ✧ A **translation** is applied to an object by repositioning it along a **straight-line path** one coordinate location to another.

$$\begin{aligned}x' &= x + tx \\ y' &= y + ty\end{aligned}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \end{bmatrix} + \begin{bmatrix} x \\ y \end{bmatrix}$$



2D Geometric Transformations

Rotation

- ☆ A **two-dimensional rotation** is applied to an object by repositioning it along a **circular path** in the xy plane.
- ☆ Positive values for the rotation angle define **counter clockwise**.
- ☆ Negative values for the rotation angle define **clockwise**.

$$\frac{A}{H} = \frac{x}{r} = \cos\phi$$

$$\frac{O}{H} = \frac{y}{r} = \sin\phi$$

$$x = r \cos \phi$$

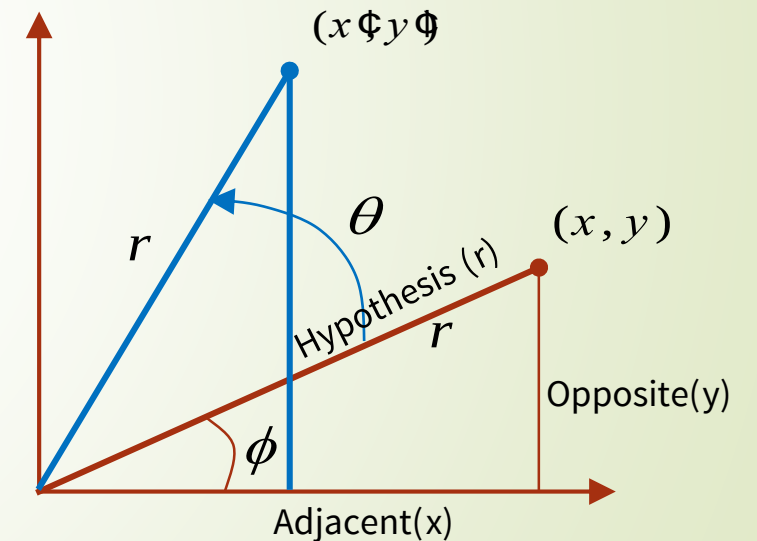
$$y = r \sin \phi$$

$$x' = r \cos(\phi + \theta)$$

$$y' = r \sin(\phi + \theta)$$

$$\begin{aligned} x' = r \cos(\phi + \theta) &= r \cos\phi \cos\theta - r \sin\phi \sin\theta \\ &= x \cos\theta - y \sin\theta \end{aligned}$$

$$\begin{aligned} y' = r \sin(\phi + \theta) &= r \sin\phi \cos\theta + r \cos\phi \sin\theta \\ &= y \cos\theta + x \sin\theta \end{aligned}$$



$$\frac{O}{H} = \sin, \frac{A}{H} = \cos, \frac{O}{A} = \tan$$

Rotation

2D Geometric Transformations

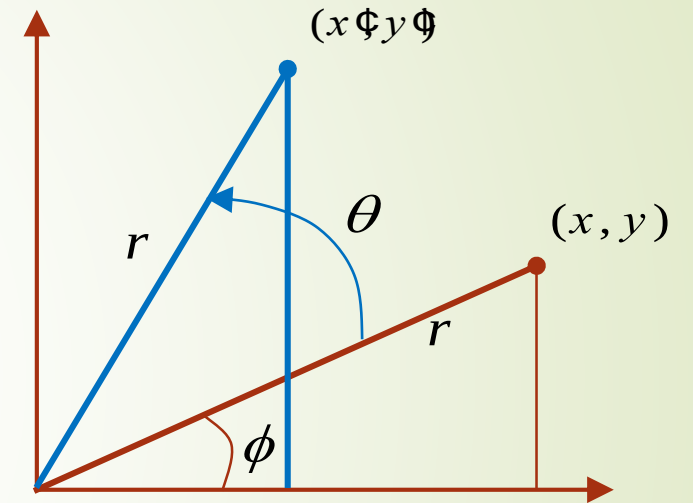
$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

$$P' = R(\theta) \cdot P$$



2D Geometric Transformations

Scaling

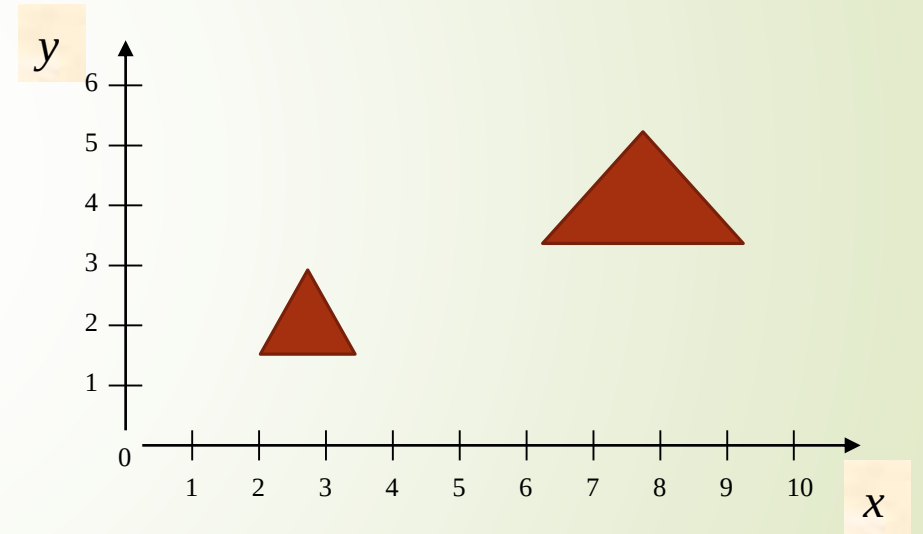
☆ A scaling transformation alters the size of an object.

$$x' = x \cdot s_x$$

$$y' = y \cdot s_y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

$$P' = S \cdot P$$



2D Geometric Transformations

Homogeneous Coordinates

- ☆ Perform a sequence of transformation such as translation followed by rotation and scaling, we need to follow a sequential process
 - ✓ Translate the coordinates,
 - ✓ Rotate the translated coordinates, and then
 - ✓ Scale the rotated coordinates to complete the composite transformation.
- ☆ To combine these three transformations into a single transformation, homogeneous coordinates are used.

Matrix Representations and Homogeneous Coordinates

☆ Translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad [P' = T(t_x, t_y) \cdot P]$$

☆ Rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad [P' = R(\theta) \cdot P]$$

☆ Scaling

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad [P' = S(s_x, s_y) \cdot P]$$

Composite Transformations

- ☆ A matrix can be set up for any **sequence of transformations** as a composite transformation matrix by calculating the *matrix product of the individual transformations*.

Translation

- ☆ Two successive translations are **additive**.

$$P' = T(tx_2, ty_2). \{T(tx_1, ty_1). P\}$$

$$= \{T(tx_2, ty_2). T(tx_1, ty_1)\}. P$$

$$T(tx_2, ty_2). T(tx_1, ty_1) = T(tx_1 + tx_2, ty_1 + ty_2)$$

$$T(tx_2, ty_2). T(tx_1, ty_1) = \begin{bmatrix} 1 & 0 & tx_2 \\ 0 & 1 & ty_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & tx_1 \\ 0 & 1 & ty_1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & tx_1 + tx_2 \\ 0 & 1 & ty_1 + ty_2 \\ 0 & 0 & 1 \end{bmatrix}$$

Composite Transformations

Rotation

☆ Two successive rotations are **additive**.

$$P' = R(\theta_2) \cdot \{R(\theta_1) \cdot P\}$$

$$= \{R(\theta_2) \cdot R(\theta_1)\} \cdot P$$

$$\mathbf{R}(\theta_2) \cdot \mathbf{R}(\theta_1) = \mathbf{R}(\theta_1 + \theta_2)$$

$$\begin{aligned} \mathbf{R}(\theta_2) \cdot \mathbf{R}(\theta_1) &= \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 \\ \sin\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_2\cos\theta_1 - \sin\theta_2\sin\theta_1 & -\cos\theta_2\sin\theta_1 - \sin\theta_2\cos\theta_1 & 0 \\ \sin\theta_2\cos\theta_1 + \cos\theta_2\sin\theta_1 & -\sin\theta_2\sin\theta_1 + \cos\theta_2\cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 \\ \sin(\theta_1 + \theta_2) & \cos(\theta_2 + \theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{R}(\theta_1 + \theta_2) \end{aligned}$$

Composite Transformations

Scaling

☆ Two successive scaling operations are **multiplicative**.

$$P' = S(sx_2, sy_2) \cdot \{S(sx_1, sy_1) \cdot P\}$$

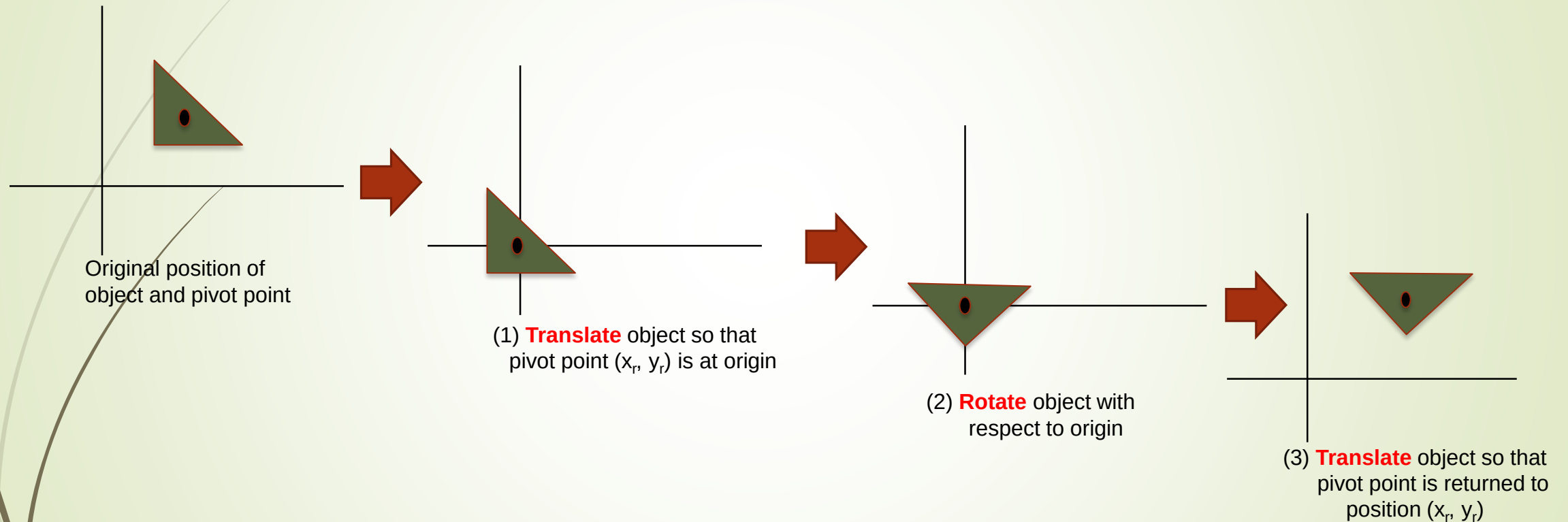
$$= \{S(sx_2, sy_2) \cdot \{S(sx_1, sy_1) \cdot P\}$$

$$\mathbf{S(sx_2, sy_2) \cdot \{S(sx_1, sy_1) = S(sx_1 \cdot sx_2, sy_1 \cdot sy_2)}$$

$$\begin{aligned} \mathbf{S(sx_2, sy_2) \cdot \{S(sx_1, sy_1)} &= \begin{bmatrix} sx_2 & 0 & 0 \\ 0 & sy_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} sx_1 & 0 & 0 \\ 0 & sy_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} sx_1 \cdot sx_2 & 0 & 0 \\ 0 & sy_1 \cdot sy_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \mathbf{S(sx_1 \cdot sx_2, sy_1 \cdot sy_2)} \end{aligned}$$

Composite Transformations

General Pivot-Point Rotation ($R(x_r, y_r, \theta)$)



$$R(x_r, y_r, \theta) = T(x_r, y_r) \cdot R(\theta) \cdot T(-x_r, -y_r)$$

Composite Transformations

☆ General Pivot-Point Rotation

$$R(x_r, y_r, \theta) = T(x_r, y_r) \cdot R(\theta) \cdot T(-x_r, -y_r)$$

$$\begin{bmatrix} 1 & 0 & x_r \\ 0 & 1 & y_r \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_r \\ 0 & 1 & -y_r \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & x_r(1 - \cos\theta) + y_r\sin\theta \\ \sin\theta & \cos\theta & y_r(1 - \cos\theta) - x_r\sin\theta \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = [R(x_r, y_r, \theta)] \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & x_r(1 - \cos\theta) + y_r\sin\theta \\ \sin\theta & \cos\theta & y_r(1 - \cos\theta) - x_r\sin\theta \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Composite Transformations

☆ General Pivot-Point Rotation

Example: Rotate the triangle with the vertices (20,20), (100,20), (50,50) with respect to its centroid (57,30) and rotation angle is given 30 degree.

$$R(x_r, y_r, \theta) = T(x_r, y_r) \cdot R(\theta) \cdot T(-x_r, -y_r) \quad \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & x_r(1 - \cos\theta) + y_r\sin\theta \\ \sin\theta & \cos\theta & y_r(1 - \cos\theta) - x_r\sin\theta \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} \cos\theta & -\sin\theta & x_r(1 - \cos\theta) + y_r\sin\theta \\ \sin\theta & \cos\theta & y_r(1 - \cos\theta) - x_r\sin\theta \\ 0 & 0 & 1 \end{bmatrix} &= \begin{bmatrix} \cos 30 & -\sin 30 & 57(1 - \cos 30) + 30\sin 30 \\ \sin 30 & \cos 30 & 30(1 - \cos 30) - 57\sin 30 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.87 & -0.5 & 57(1 - 0.87) + 30(0.5) \\ 0.5 & 0.87 & 30(1 - 0.87) - 57(0.5) \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.87 & -0.5 & 57(0.13) + 30(0.5) \\ 0.5 & 0.87 & 30(0.13) - 57(0.5) \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.87 & -0.5 & 22.41 \\ 0.5 & 0.87 & -24.6 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Composite Transformations

☆ General Pivot-Point Rotation

Example:

For the first point, (20,20)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0.87 & -0.5 & 22.41 \\ 0.5 & 0.87 & -24.6 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 20 \\ 20 \\ 1 \end{bmatrix} \Rightarrow ?$$

For the first point, (100,20)

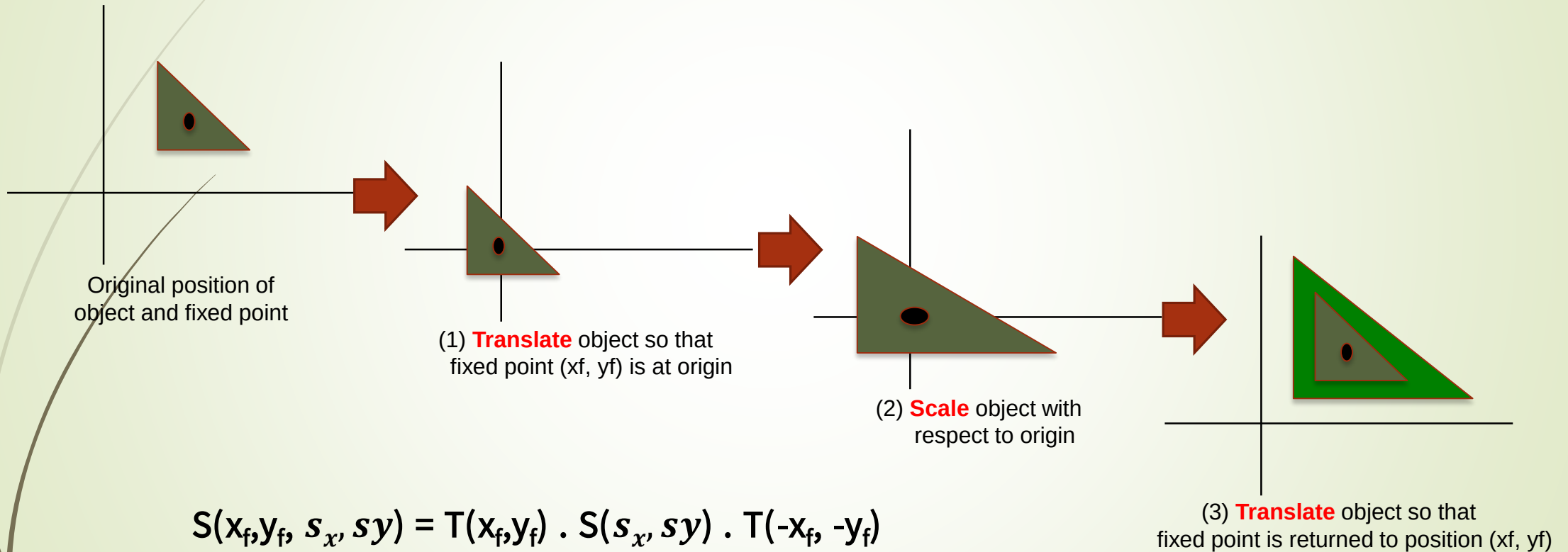
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0.87 & -0.5 & 22.41 \\ 0.5 & 0.87 & -24.6 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 100 \\ 20 \\ 1 \end{bmatrix} \Rightarrow ?$$

For the first point, (50,50)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0.87 & -0.5 & 22.41 \\ 0.5 & 0.87 & -24.6 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 50 \\ 50 \\ 1 \end{bmatrix} \Rightarrow ?$$

Composite Transformations

☆ General Fixed-Point Scaling



Composite Transformations

☆ General Fixed-Point Scaling

$$S(x_f, y_f, s_x, s_y) = T(x_f, y_f) \cdot S(s_x, s_y) \cdot T(-x_f, -y_f)$$

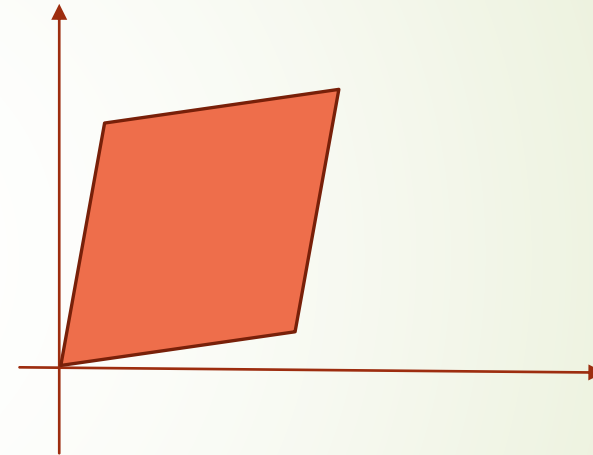
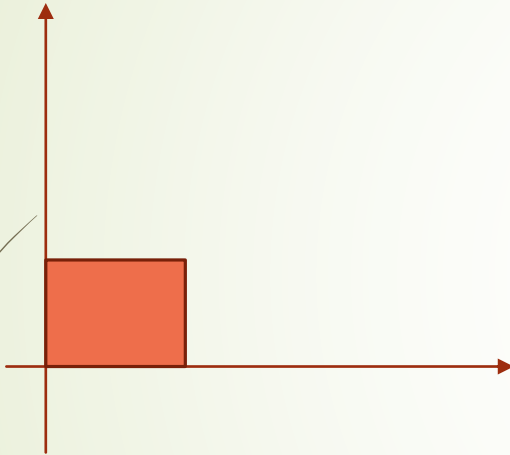
$$\begin{bmatrix} 1 & 0 & x_f \\ 0 & 1 & y_f \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_f \\ 0 & 1 & -y_f \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & x_f(1 - s_x) \\ 0 & s_y & y_f(1 - s_y) \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & x_f(1 - s_x) \\ 0 & s_y & y_f(1 - s_y) \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Example: Scale the triangle with vertices (20,20), (50,40), (80,60) by using scaling factor $s_x=0.5$ and $s_y=0.5$ with respect to its given fixed point position (30,30).

Composite Transformations

☆ General Scaling Direction



$$T_{\text{composite}} = R(-\theta) \cdot S(s_x, s_y) \cdot R(\theta)$$

Composite Transformations

☆ General Scaling Direction

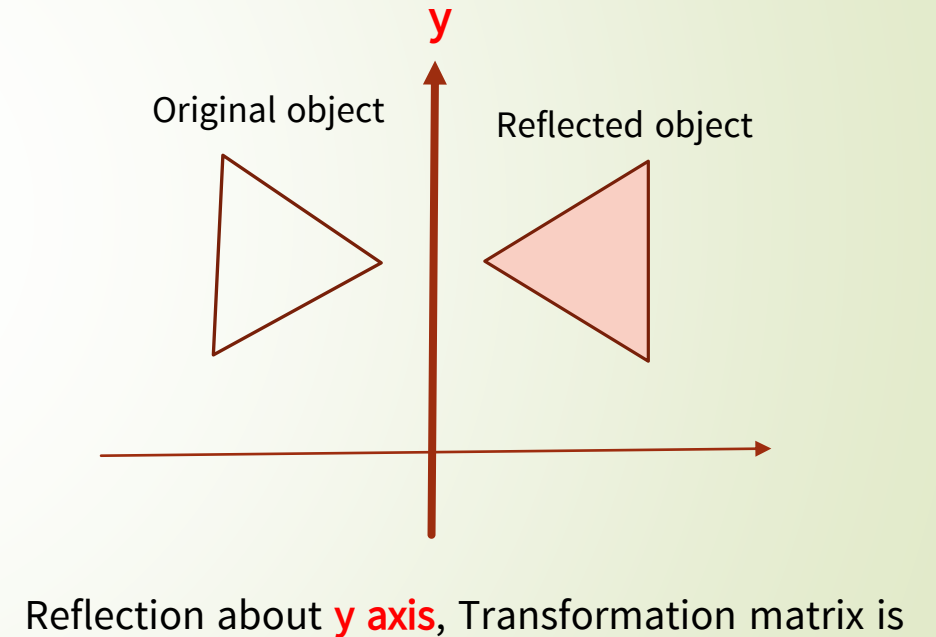
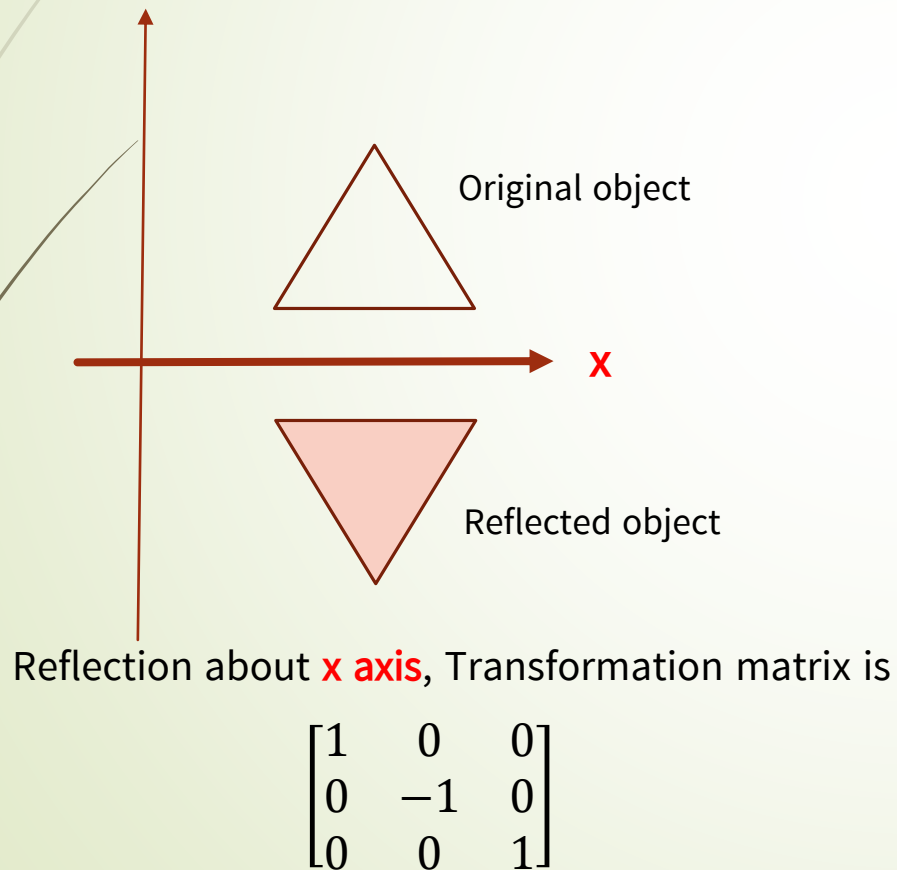
$$\begin{aligned}
 R(-\theta) \cdot S(s_x, s_y) \cdot R(\theta) &= \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) & 0 \\ \sin(-\theta) & \cos(-\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_x \cos\theta & -s_x \sin\theta & 0 \\ s_y \sin\theta & s_y \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} s_x \cos^2\theta + s_y \sin^2\theta & -s_x \cos\theta \sin\theta + s_y \sin\theta \cos\theta & 0 \\ -s_x \cos\theta \sin\theta + s_y \sin\theta \cos\theta & s_x \sin^2\theta + s_y \cos^2\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} s_x \cos^2\theta + s_y \sin^2\theta & (s_y - s_x) \cos\theta \sin\theta & 0 \\ (s_y - s_x) \cos\theta \sin\theta & s_x \sin^2\theta + s_y \cos^2\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x \cos^2\theta + s_y \sin^2\theta & (s_y - s_x) \cos\theta \sin\theta & 0 \\ (s_y - s_x) \cos\theta \sin\theta & s_x \sin^2\theta + s_y \cos^2\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Other Transformations

Reflection

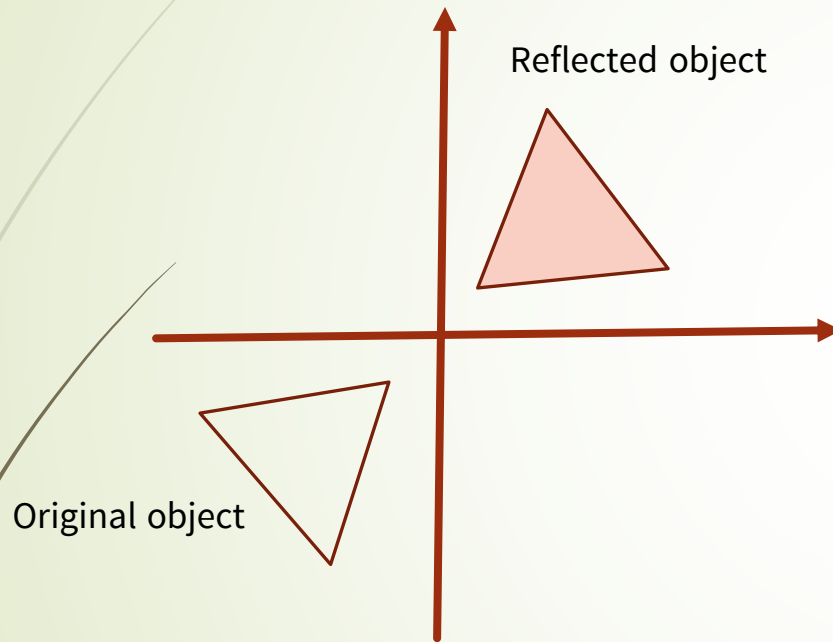
- ☆ Reflection is a transformation that produces a mirror image of an object.
- ☆ It is obtained by rotating the object by 180 degree about the reflection axis.



$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

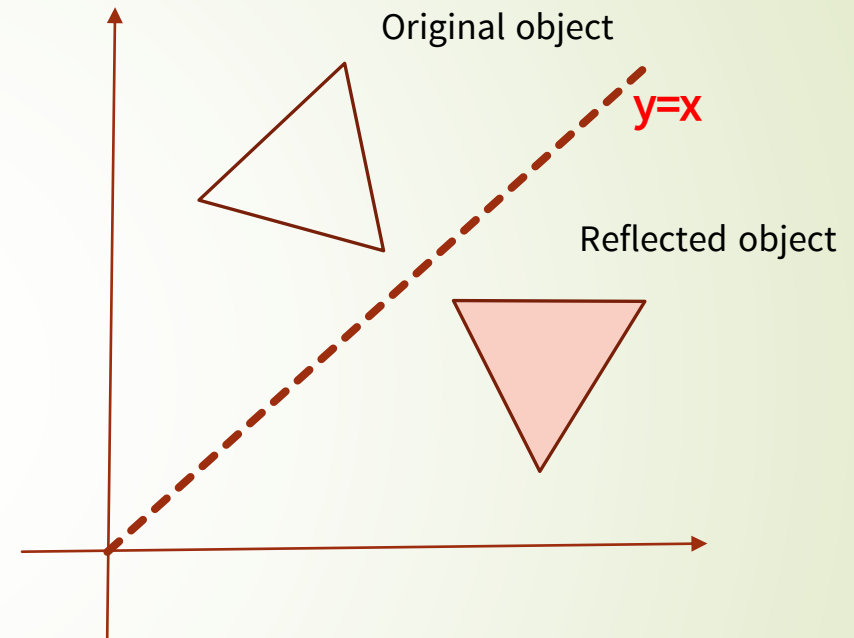
Other Transformations

Reflection



Reflection through **coordinate origin**, Transformation matrix is

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



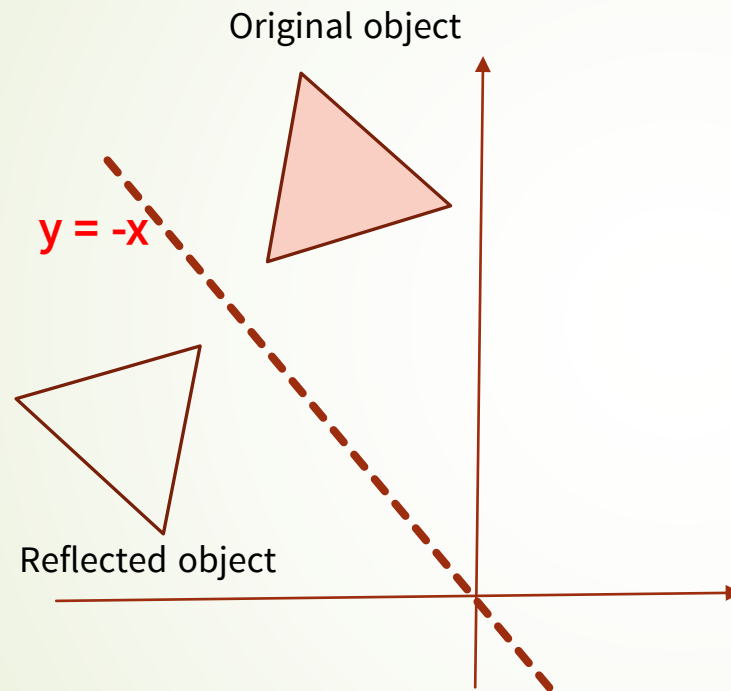
Reflection with respect to a line **y=x**,
Transformation sequence is

$R(45) \cdot \text{Reflection}(x\text{-axis}) \cdot R(-45)$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Other Transformations

Reflection



Reflection with respect to a line $y = -x$,
Transformation sequence is

$R(45).Reflection(y\text{-axis}).R(-45)$

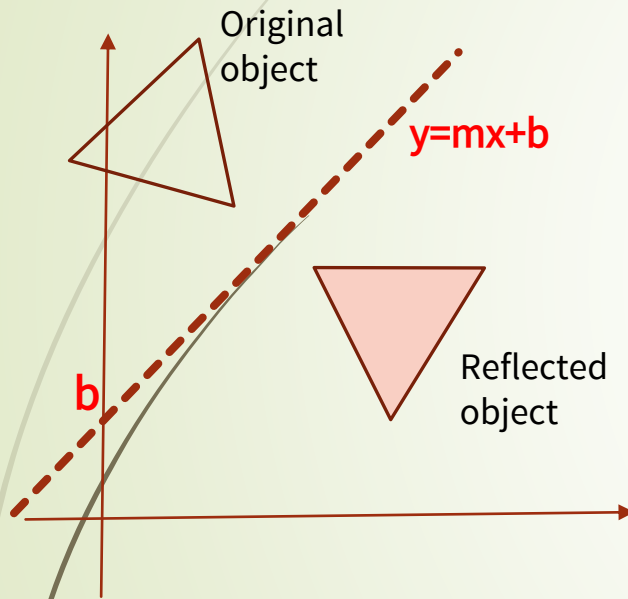
$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Other Transformations

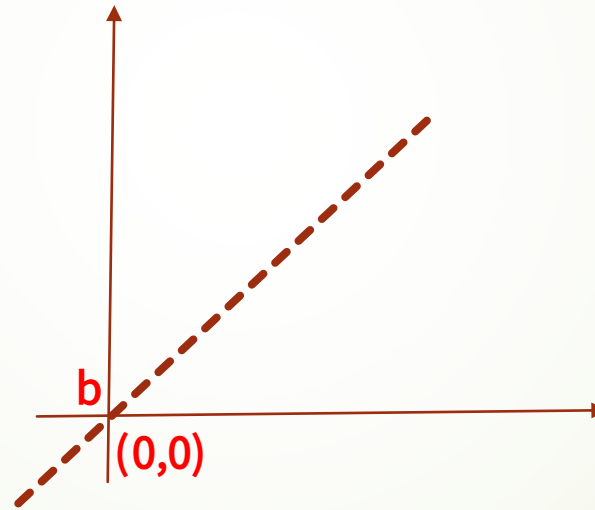
- ☆ Example: Find the reflection about the X-axis (reflection 180 deg) for the triangle (2,2),(4,2), (3,4).
- ☆ Example: Find the reflection about $y=x$ axis (reflection $\theta = 45$).

Other Transformations

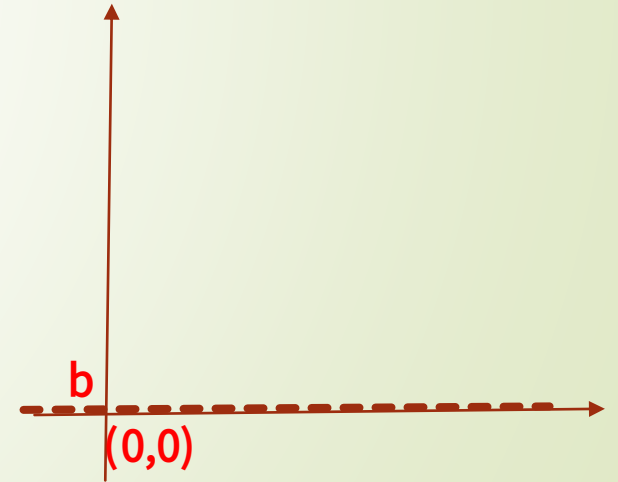
Reflection with respect to a line $y=mx+b$



1. Translate $T(-t_x, -t_y)$

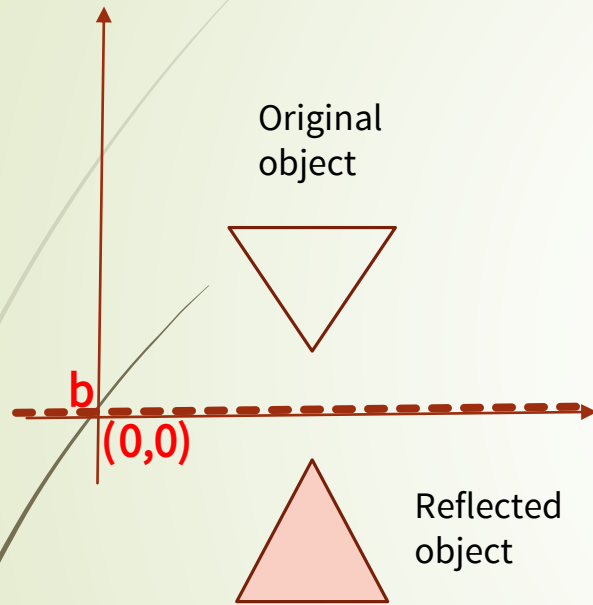


2. Rotate $R(-\theta)$

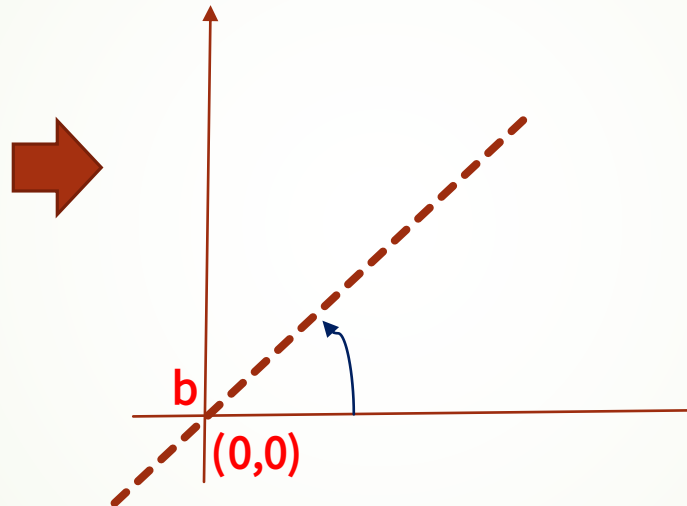


Other Transformations

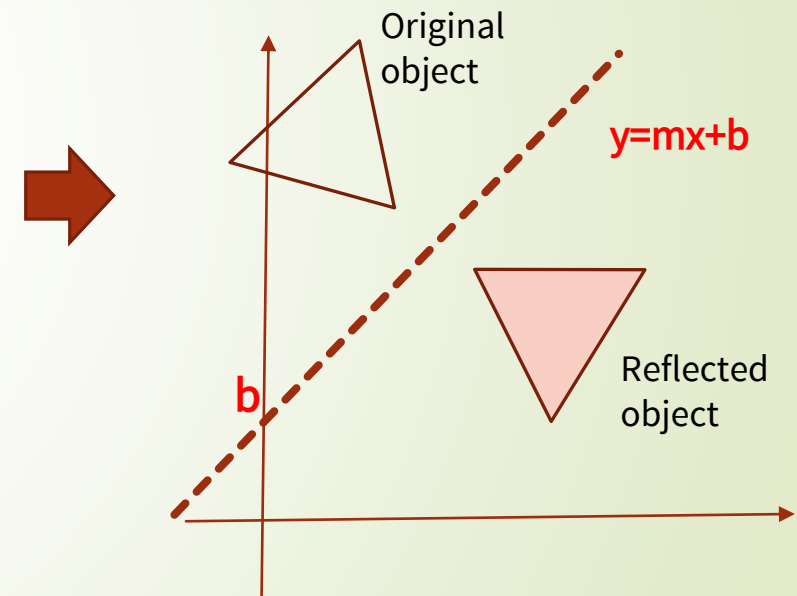
3. Reflection X-axis



4. Rotate $R(\theta)$



5. Translate $T(t_x, t_y)$

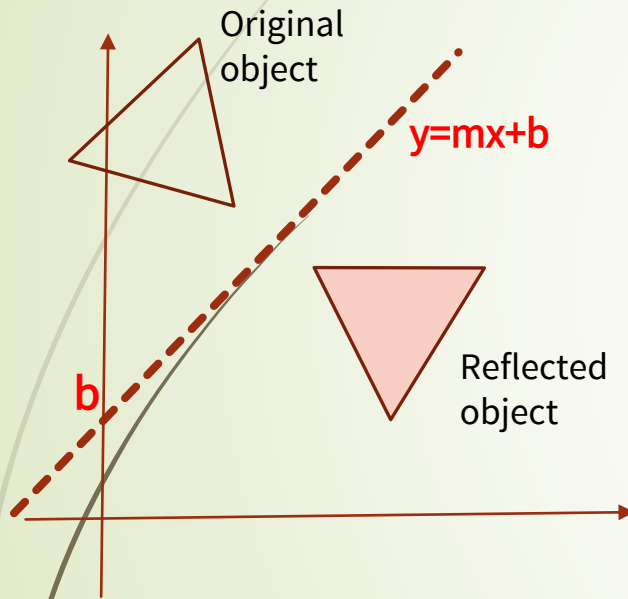


Reflection with respect to a line $y=mx+b$, Transformation sequence is

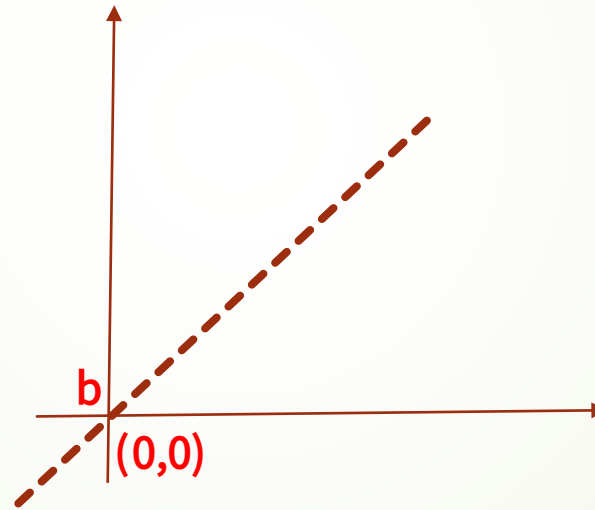
$T(t_x, t_y)$. $R(\theta)$. Reflection on X-axis. $R(-\theta)$. $T(-t_x, -t_y)$

Other Transformations

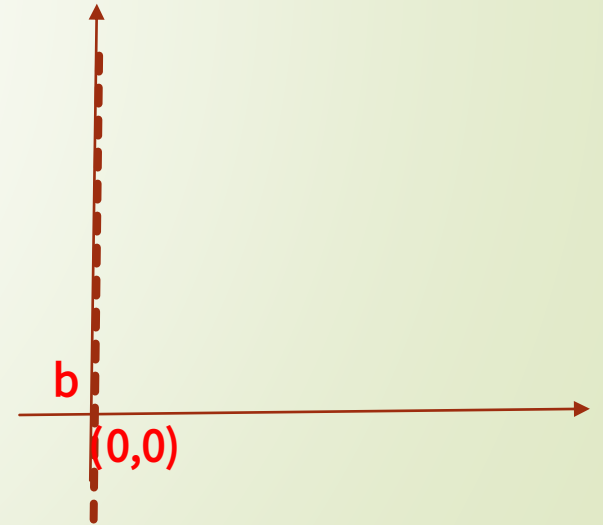
Reflection with respect to a line $y=mx+b$



1. Translate $T(-t_x, -t_y)$

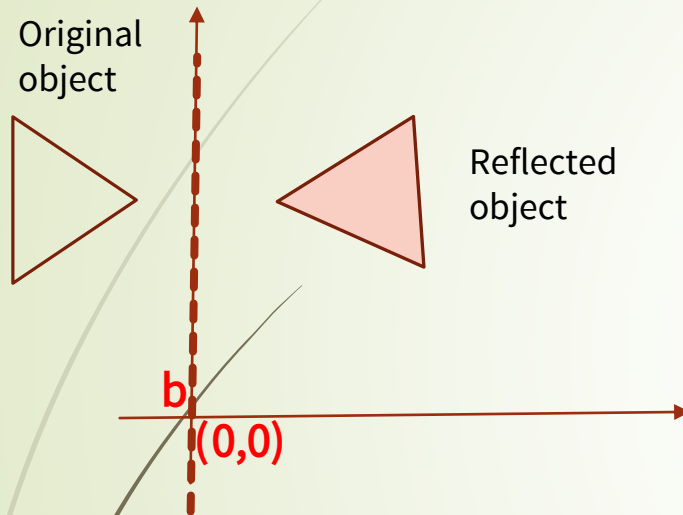


2. Rotate $R(\theta)$

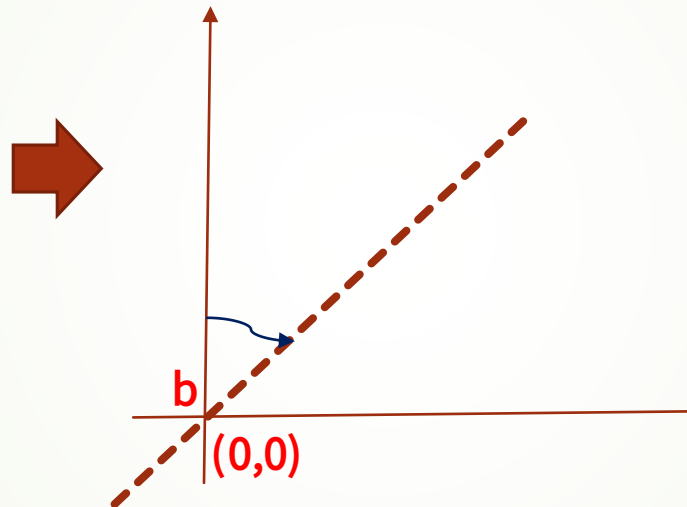


Other Transformations

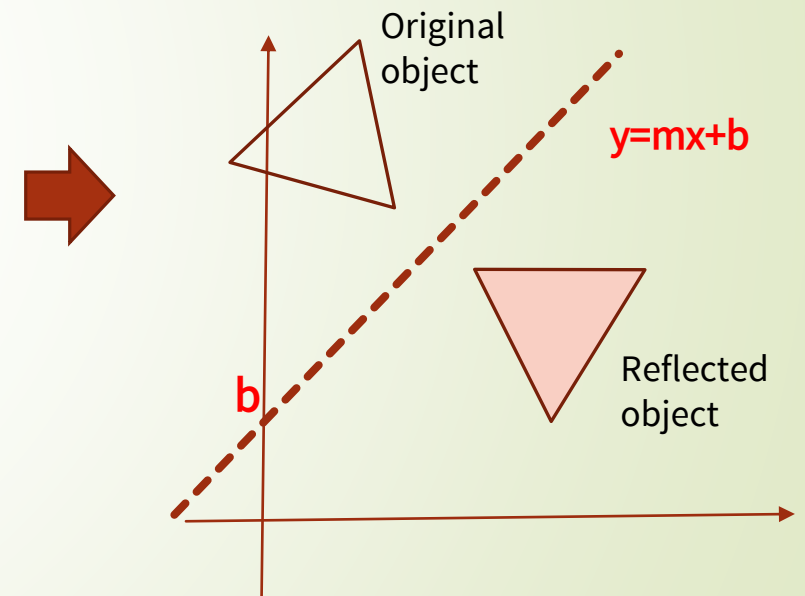
3. Reflection Y-axis



4. Rotate $R(-\theta)$



5. Translate $T(t_x, t_y)$

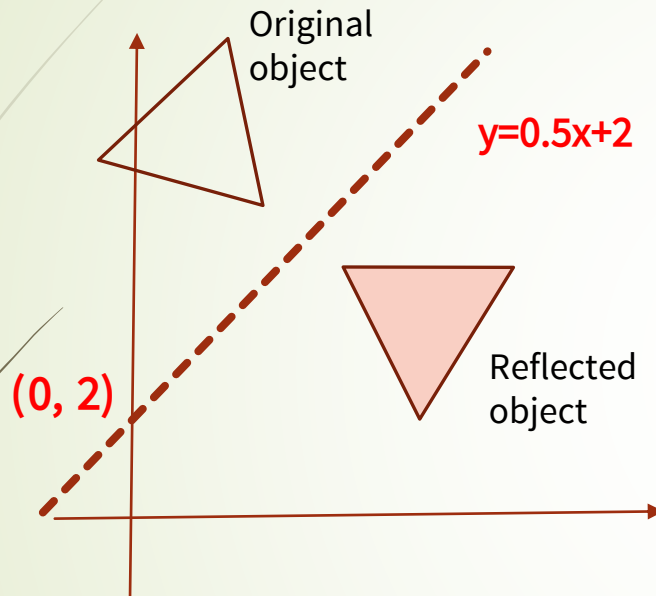


Reflection with respect to a line $y=mx+b$, Transformation sequence is

$T(t_x, t_y)$, $R(-\theta)$, Reflection on Y-axis, $R(\theta)$, $T(-t_x, -t_y)$

Other Transformations

Reflection with respect to a line $y=mx+b$

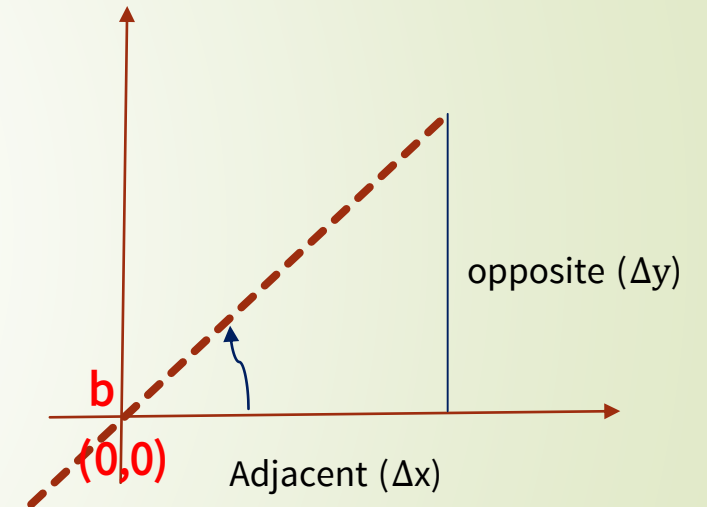


$$y = 0.5x+2$$

$$x=0, y=2$$

$$x=1, y=2.5$$

Rotate $R(-\theta)$



$$\tan \theta = \frac{\text{opposite}(\Delta y)}{\text{Adjacent}(\Delta x)} = m$$

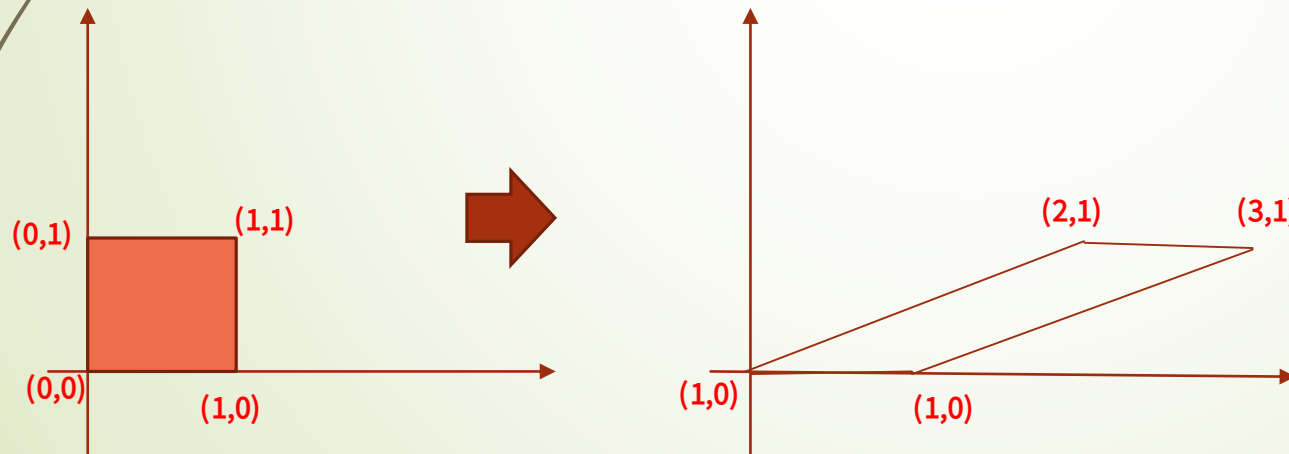
$$\theta = \tan^{-1} m$$

Other Transformations

Shearing

- ☆ A transformation that distorts the shape of an object.
- ☆ Two common shearing transformations are :
 - Shift coordinate x values
 - Shift coordinate y values

An x-direction shear relative to the x axis



$$\begin{aligned}x' &= x + sh_x \cdot y, \\ y' &= y\end{aligned}$$

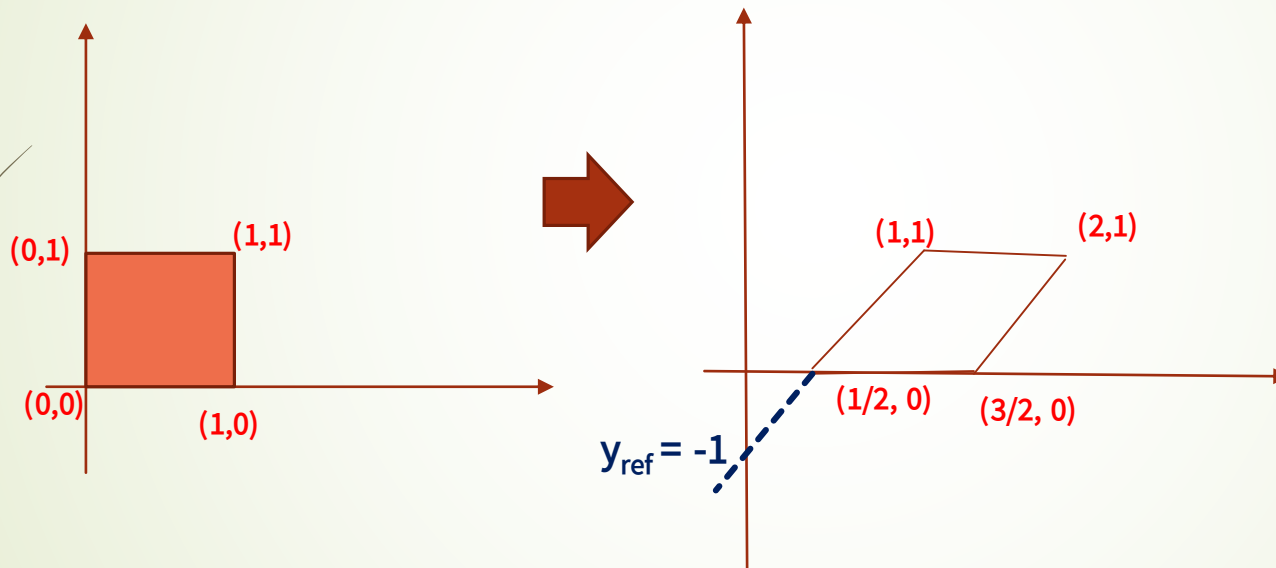
Transformation matrix is

$$\begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Other Transformations

Shearing

An x-direction shears relative to other reference line $y = y_{ref}$



$$\begin{aligned}x' &= x + sh_x \cdot (y - y_{ref}), \\y' &= y\end{aligned}$$

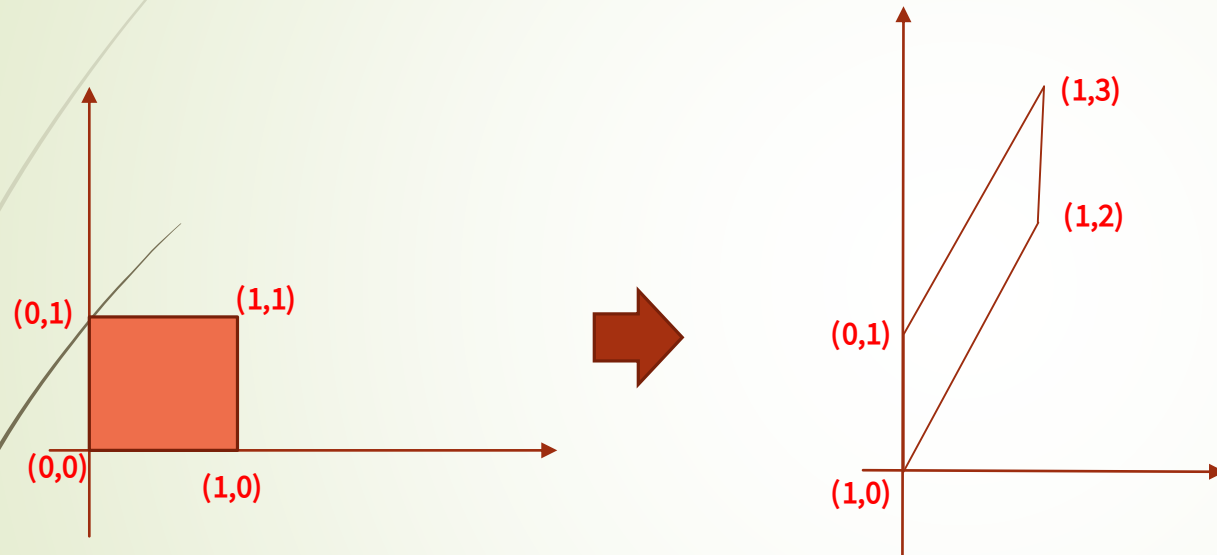
Transformation matrix is

$$\begin{bmatrix} 1 & sh_x & -sh_x \cdot y_{ref} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Other Transformations

Shearing

A y-direction shear relative to the y axis



$$\begin{aligned}x' &= x, \\ y' &= y + sh_y \cdot x\end{aligned}$$

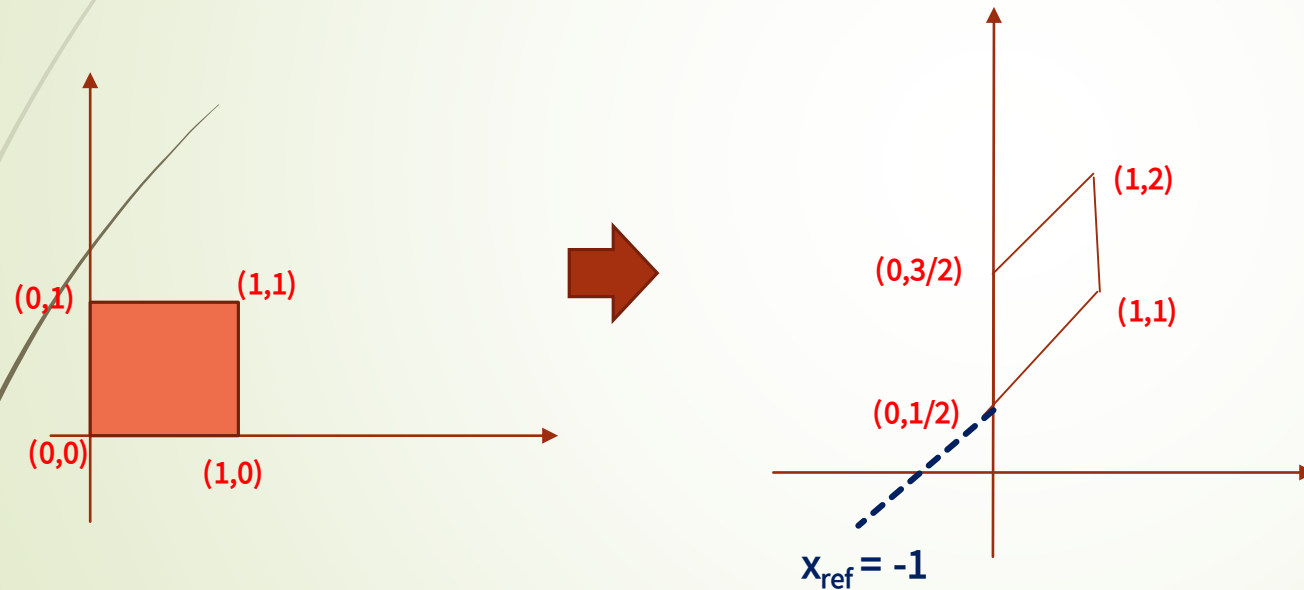
Transformation matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Other Transformations

Shearing

A y-direction shearing relative to other reference line $x = x_{\text{ref}}$



$$\begin{aligned} x' &= x, \\ y' &= y + sh_y \cdot (x - x_{\text{ref}}) \end{aligned}$$

Transformation matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & -sh_y \cdot x_{\text{ref}} \\ 0 & 0 & 1 \end{bmatrix}$$

Exercises

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- ✧ Prove that the multiplication of transformation matrices (or each of the following sequence of operations) is commutative:
 - (a) Two successive rotations.
 - (b) Two successive translations.
 - (c) Two successive scalings.
- ✧ Show that transformation matrix, for a reflection about the line $y = x$, is equivalent to a reflection relative to the x axis followed by a counterclockwise rotation of 90° .
- ✧ Show that transformation matrix, for a reflection about the line $y = -x$, is equivalent to a reflection relative to the y axis followed by a counterclockwise rotation of 90° .
- ✧ Show that two successive reflections about either of the coordinate axes is equivalent to a single rotation about the coordinate origin.
- ✧ Show that two successive reflections about any line passing through the coordinate origin is equivalent to a single rotation about the origin.